Method for Walk Correction using Leading Edge Slope:

Derived from dual threshold feature of TOFHIR2C

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Abstract

Precision timing at the HL-LHC requires rapid, repeatable calibration as detector response evolves. With many interactions per bunch crossing, an external particle-arrival time reference is not readily available. The CMS barrel timing layer (BTL) therefore motivates calibration procedures based on internal measurements and a consistent parametrization across about 330 000 channels.

The September 2025 test beam exposed BTL sensors with TOFHIR2C readout to 180 GeV pions. Separate acquisitions of the same reference module provide dual-threshold slopes and MCP-referenced timing; they contain different events. Extending an earlier bench study, the mean slope versus charge defines an inverse-slope walk correction with one amplitude parameter, AWC, per SiPM and configuration.

Comparison with independent microchannel-plate (MCP) reference calibrations shows good agreement between the correction shapes. Internal charge and slope information provide candidate predictors of AWC, with the MCP results serving as benchmarks. Predictive precision and transfer to other operating conditions remain to be validated for reference-free calibration.

BTL and TOFHIR2C: Refs. [3, 7]; dual-threshold motivation and bench study: Ref. [5].

Charge-based calibration works, but its coefficients vary

The MCP reference note obtains good walk descriptions with a quadratic in inverse charge [6]:

$$W_i(Q) = a_i + b_i/Q + c_i/Q^2.$$

Bar	SiPM	a_i [ns]	b_i [ns ADC]	c_i [10^4 ns ADC^2]
7	Left	0.080 ± 0.067	238.3 ± 35.8	-2.16 ± 0.47
7	Right	-0.515 ± 0.090	422.6 ± 68.0	-5.58 ± 1.27
8	Left	0.122 ± 0.027	198.5 ± 13.1	-1.52 ± 0.16
8	Right	-0.229 ± 0.033	388.9 ± 26.2	-5.13 ± 0.51

With Q expressed as raw, uncalibrated ADC values, the shape coefficients vary substantially across these four SiPMs.

Original unweighted fits; 1σ errors assume independent profile means, fixed bins and selection. No systematics; coefficients strongly correlated. Q is not gain-calibrated; a_i includes an arbitrary offset. Coefficient variation does not measure calibration precision. Earlier experience: ALICE [1, 2].

TOFHIR2C measures the leading-edge slope with two thresholds

Larger pulses cross a fixed threshold earlier: the timestamp depends on amplitude [4]. Two-threshold timing has earlier implementations [8, 9]; TOFHIR2C provides both crossings [7, 5].

$$I_1 < I_2, \qquad t_1 < t_2, \qquad S = \frac{I_2 - I_1}{t_2 - t_1} \quad [\mu\mathrm{A/ns}].$$

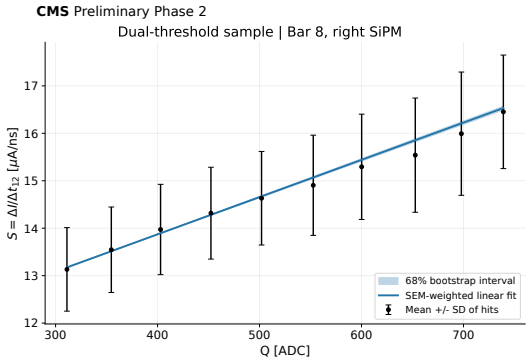
Locally, an approximately linear leading edge gives

$$\Delta t \simeq \frac{\Delta I}{S}.$$

Physical model: describe amplitude walk with an inverse slope, rather than an independent polynomial shape for every channel.

S is a finite-interval slope proxy, not the instantaneous derivative. The dual-threshold sample uses $\Delta I = 7.8125\,\mu\mathrm{A}$ from the implemented threshold conversion. The fitted AWC need not equal this threshold separation. GitHub: timing and threshold reconstruction [11].

Measure the charge-to-slope mapping using internal hits



Typical single-hit relative uncertainties: Direct slope: $\sim 15\%$; charge: $< 5\%$. Use the more precise Q to estimate the mean slope:

$$\overline{S}_i(Q) = c_{0,i} + c_{1,i} Q.$$

Ten means; weights $1/\mathrm{SEM}^2$; no MCP time.

Reference module, bar 8, right SiPM: 19 312 hits. Both SiPMs required; other module unconstrained. Physical crossings; peak-right charge cut. Points: means; bars: hit standard deviation (SD). Standard error of the mean: $\mathrm{SEM} = \mathrm{SD}/\sqrt{N}$, for N hits per bin. Band: conditional 68% interval, 1000 hit-bootstrap refits, fixed bins.

One amplitude describes walk once the slope is calibrated

$$W(Q) = \frac{\mathrm{AWC}}{\overline{S}(Q)}, \qquad t_{\mathrm{corr}} = t_{\mathrm{measured}} - W(Q).$$

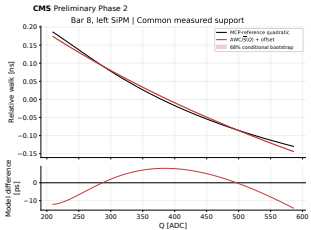
- Internal hits determine $\overline{S}(Q)$, the mean slope at a given charge, for each SiPM.
- One parameter, AWC (in μA), sets the walk magnitude.
- AWC is constant per SiPM and configuration; the correction varies with Q.
- Channel timing offsets are a separate calibration.

The remaining task is to predict AWC from internal observables. Use the MCP-calibrated channels to develop and test that prediction.

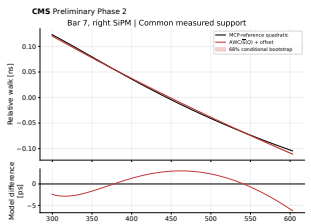
Apply the model within its calibrated charge range, with $\overline{S}(Q) > 0$. The slope fit alone does not fix AWC. The common functional form follows the dual-threshold model in Ref. [5].

Obtain benchmark AWC by matching the MCP curves

Bar 8, left SiPM



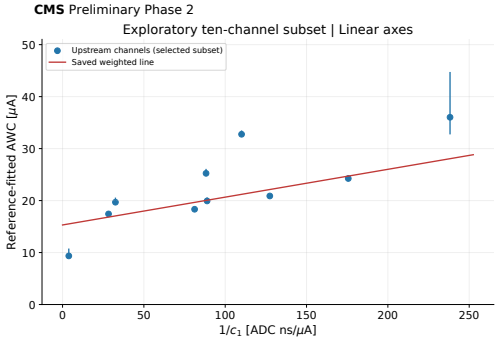
Bar 7, right SiPM



$$(\mathrm{AWC}_{\mathrm{fit}}, b_{\mathrm{fit}}) = \arg\min_{\mathrm{AWC},\, b} \sum_{Q_j \in \Omega} \left[f_{\mathrm{MCP}}(Q_j) - \left(\frac{\mathrm{AWC}}{\overline{S}(Q_j)} + b \right) \right]^2 .$$

Black: MCP quadratic f_{MCP} [6]; red: AWC model; b: free offset. Ω: equally weighted Q grid in common populated intervals; no gaps or extrapolation. Independent reference-module samples. Bar 8, left SiPM: 6.80 ps RMS mismatch. Lower panels: curve differences, not resolution. Bands: slope-fit statistics, MCP curve fixed; clock alignment: LOWESS [10]. AWC is fitted to MCP here, not predicted by the following candidates.

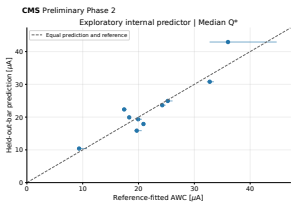
Candidate 1: AWC versus inverse slope coefficient



An exploratory trend: AWC versus $1/c_1$, the inverse slope-to-charge coefficient. Each point is one SiPM, not a charge bin. AWC is benchmarked against MCP; c_1 uses internal hits. The ten-SiPM trend motivates prediction, but the line is not a precision calibration ($\chi^2/\mathrm{ndf} = 538.6/8$). Bars: slope-only 16–84% bootstrap intervals; reference errors and correlations are omitted. More independent data are needed.

Candidate 2: a physics-motivated AWC predictor

Physics motivation: similar scintillation pulses and front-end shaping motivate testing a shared timing sensitivity to the same fractional charge change.



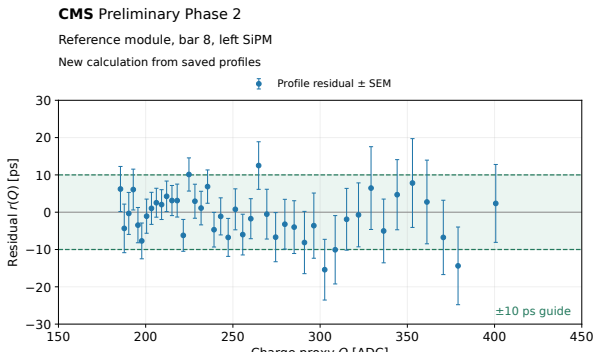
$$K \simeq -\left.\frac{dW}{d\ln Q}\right|_{Q_*}, \quad \mathrm{AWC}_{\mathrm{pred}} = \frac{K\overline{S}(Q_*)^2}{c_1 Q_*}.$$

Required assumptions

1. $W = \mathrm{AWC}/\overline{S}$ holds locally, with constant AWC per SiPM.
2. The fitted $\overline{S}(Q) = c_0 + c_1 Q$ is reliable near Q_*.
3. Q_*, the selected-charge median per SiPM, samples comparable pulse regions.
4. Similar pulses and operating conditions allow a common K across SiPMs and bars.

Leave-one-bar-out test: ten SiPMs, six bars; K learned from other bars' MCP AWC. RMS: 3.22 μA (14.1%). Dashed: equality. Bars: conditional reference-AWC intervals, not prediction intervals. Full uncertainties and correlations unvalidated. Transfer across thresholds, overvoltage or charge-integration settings untested; recalibration may be needed.

The AWC correction describes the walk at the 10 ps scale



- **Y axis:** mean time minus MCP, the AWC correction and a constant offset.
- **43 charge bins:** within the slope calibration range, with 6.1 ps RMS of bin residuals.
- ± 10 **ps guide:** indicates the residual-walk scale, not an all-point bound.

$r(Q) = \langle t_{\mathrm{thr1}} - t_{\mathrm{MCP}} \rangle_Q - \mathrm{AWC}/\overline{S}(Q) - b$; b: fitted constant. New MCP-profile calculation [6]: saved $\overline{S}(Q)$, fixed $\mathrm{AWC} = 20\,\mu\mathrm{A}$. Bars: input profile-mean SEM, conditional on fixed correction and offset. Guide: not a confidence band. This is neither single-hit resolution nor AWC-prediction validation.

References

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